INTERNSHIP | PROJECT REPORT

CYBERSECURITY &

ETHICAL HACKING

Task-2: Network Security and Scanning

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ADDIS ABABA, ETHIOPIA

October 2026

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Nmap Scan Report and OpenVAS Vulnerability Assessment

1. Executive Summary

This report presents the results of network reconnaissance and vulnerability-assessment activities conducted in an authorized virtual lab for educational purposes. The assessment used Kali Linux to scan the Metasploitable-2 target at 10.0.0.6 within the 10.0.0.0/24 lab network.

Nmap was used to discover active hosts, identify open ports, detect service names and versions, estimate the target operating system, and enumerate SMB user accounts. The scan identified numerous exposed services, including FTP, SSH, Telnet, SMTP, HTTP, SMB, MySQL, PostgreSQL, VNC, and other network services. Several services were identified as running legacy software versions that require further security review.

The report also includes evidence from the OpenVAS vulnerability-assessment interface. The findings provide a basis for identifying potential security weaknesses and recommending remediation. However, detected services and version information alone do not establish that a system is exploitable; findings should be validated and prioritized before remediation.

2. Objective

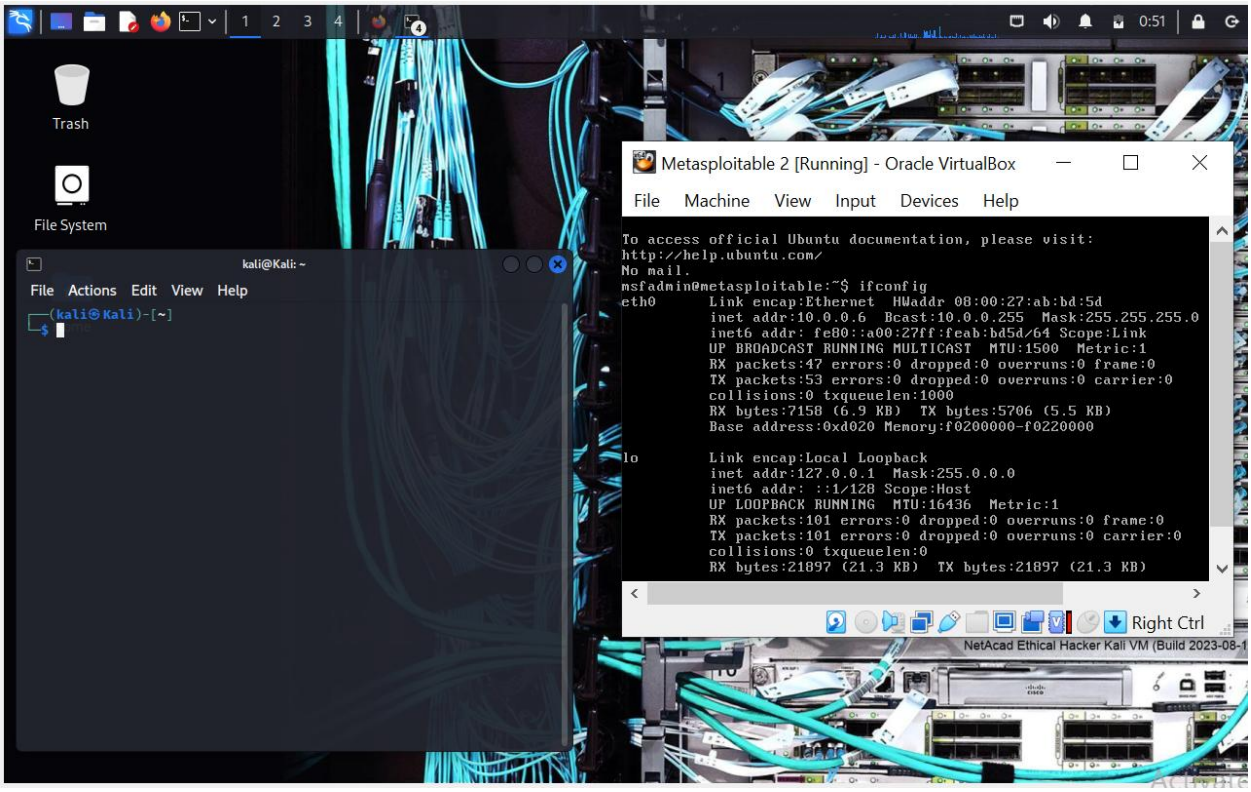
The primary objective of this assessment was to develop practical skills in network reconnaissance, service enumeration, and vulnerability assessment within a controlled virtual lab.

The specific objectives were to:

- Discover active hosts on the lab network.
- Identify open TCP ports and associated services.
- Determine service names and software versions using Nmap.
- Estimate the target operating system.

- Practice TCP SYN and FIN scanning techniques.
- Enumerate SMB user accounts to understand information exposure.
- Review the OpenVAS vulnerability-assessment interface and available findings.
- Identify potential security weaknesses and recommend appropriate mitigation measures.

3. Screenshot of Kali Linux and metasploitable-2



Nmap Scan Report

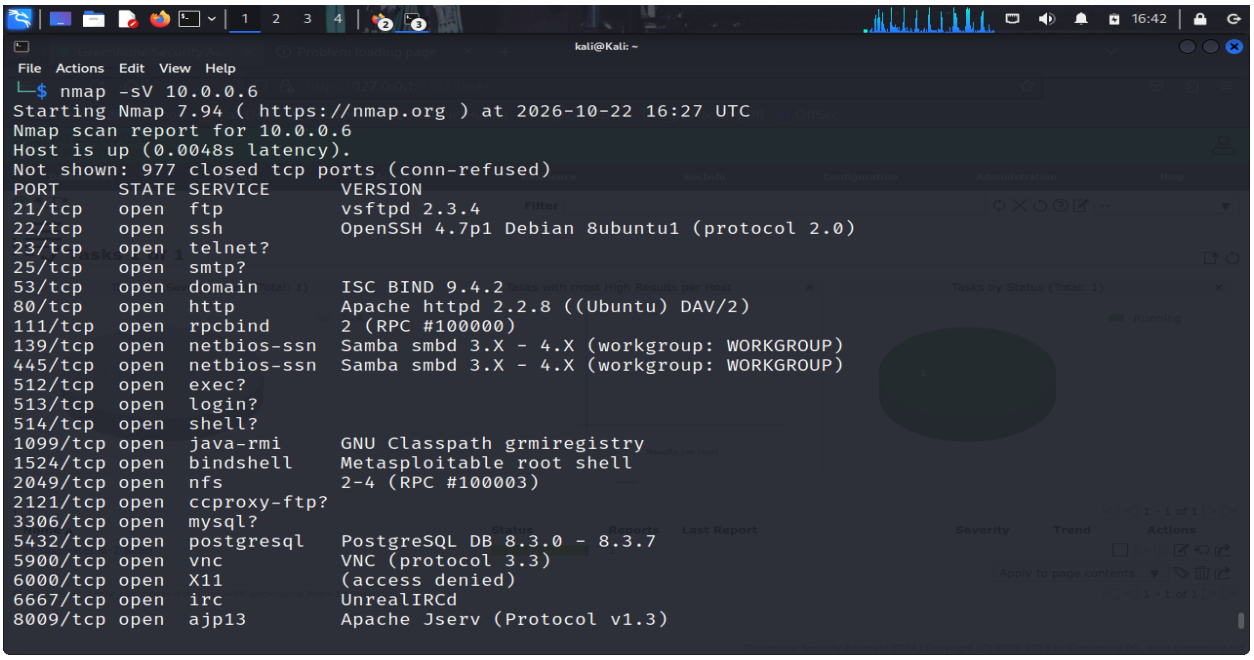
4. Methodology

The assessment was conducted using Kali Linux in a VirtualBox environment, targeting the Metasploitable-2 machine at 10.0.0.6.

The following tools and techniques were used:

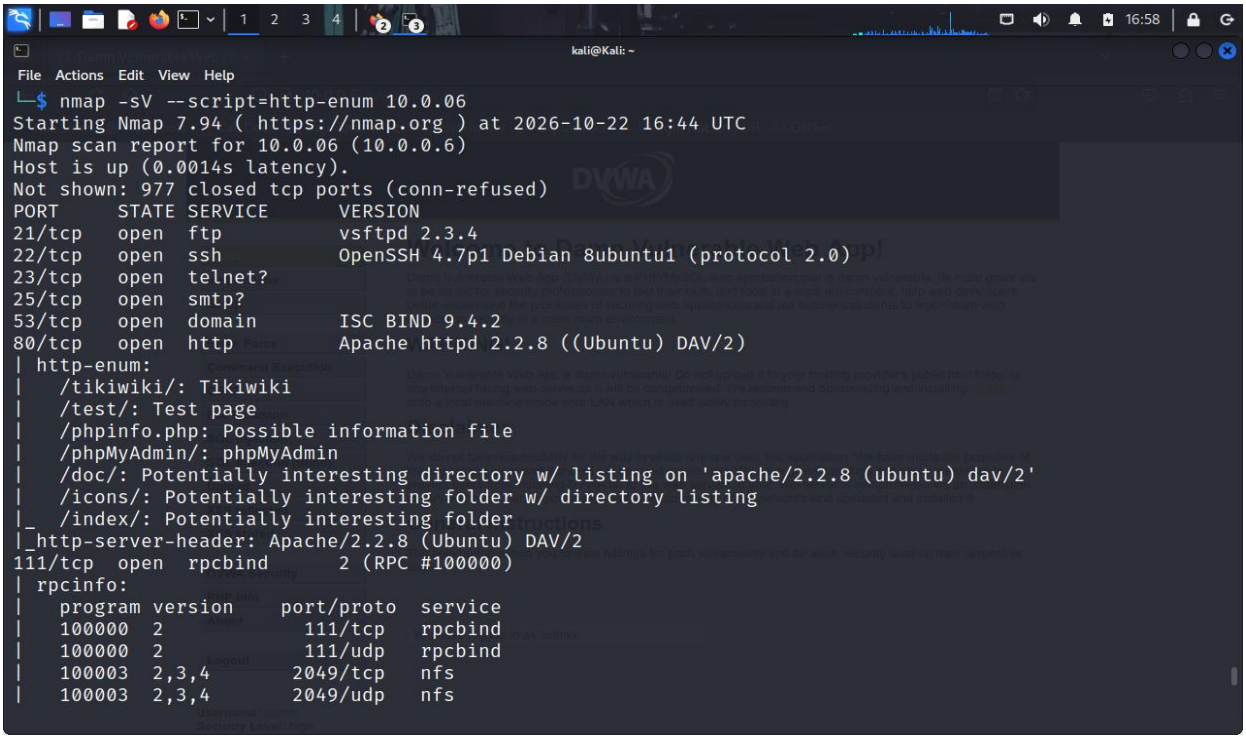
4.1. Service Enumeration

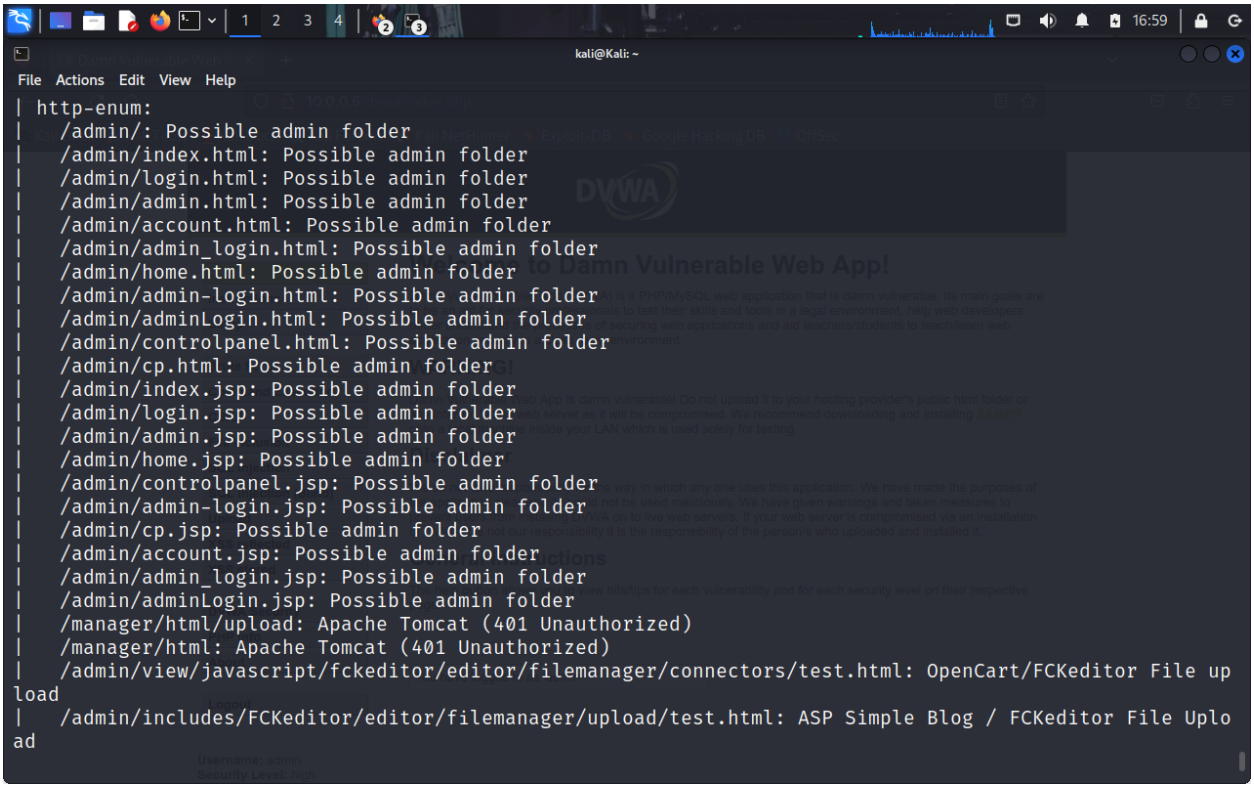
nmap –sV 10.0.0.6



4.2. Web Service Enumeration

The command nmap -sV --script=http-enum 10.0.0.6 was included to investigate discoverable web resources and directories.





4.3. Host Discovery

The command nmap -sn 10.0.0.0/24 was used to identify active hosts on the lab network.

(kali㉿Kali)-[~]

└─$ nmap -sn 10.0.0.0/24

Starting Nmap 7.94 (https://nmap.org) at 2026-10-21 01:00 UTC

Nmap scan report for 10.0.0.2

Host is up (0.00067s latency).

Nmap scan report for 10.0.0.6

Host is up (0.0010s latency).

Nmap done: 256 IP addresses (2 hosts up) scanned in 15.83 seconds

4.4. Timing Options (-T0 to –T5)

eg adjusting the speed –T0 is slowest an –T5 is fastest but more detectable by security tools

sudo nmap –T4 10.0.0.6

4.5. Port Scanning

The command sudo nmap -sS 10.0.0.6 was used to identify open TCP ports using a TCP SYN scan. A FIN scan was also performed to observe how the target responded to different TCP probes.

┌──(kali㉿Kali)-[~]

└─$ sudo nmap -sS 10.0.0.6

[sudo] password for kali:

Starting Nmap 7.94 (https://nmap.org) at 2026-10-21 00:53 UTC

Nmap scan report for 10.0.0.6

Host is up (0.00047s latency).

Not shown: 977 closed tcp ports (reset)

PORT STATE SERVICE

21/tcp open ftp

22/tcp open ssh

23/tcp open telnet

25/tcp open smtp

53/tcp open domain

80/tcp open http

111/tcp open rpcbind

139/tcp open netbios-ssn

445/tcp open microsoft-ds

512/tcp open exec

513/tcp open login

514/tcp open shell

1099/tcp open rmiregistry

1524/tcp open ingreslock

2049/tcp open nfs

2121/tcp open ccproxy-ftp

3306/tcp open mysql

5432/tcp open postgresql

5900/tcp open vnc

6000/tcp open X11

6667/tcp open irc

8009/tcp open ajp13

8180/tcp open unknown

MAC Address: 08:00:27:AB:BD:5D (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 13.64 seconds

4.6. Sends Tcp Fin Packets to Probe Port States

┌──(kali㉿Kali)-[~]

└─$ sudo nmap -sF 10.0.0.6 Nmap scan report for 10.0.0.6

Host is up (0.00050s latency).

Not shown: 977 closed tcp ports (reset)

PORT STATE SERVICE

21/tcp open|filtered ftp

22/tcp open|filtered ssh

23/tcp open|filtered telnet

25/tcp open|filtered smtp

53/tcp open|filtered domain

80/tcp open|filtered http

111/tcp open|filtered rpcbind

139/tcp open|filtered netbios-ssn

445/tcp open|filtered microsoft-ds

512/tcp open|filtered exec

513/tcp open|filtered login

514/tcp open|filtered shell

1099/tcp open|filtered rmiregistry

1524/tcp open|filtered ingreslock

2049/tcp open|filtered nfs

2121/tcp open|filtered ccproxy-ftp

3306/tcp open|filtered mysql

5432/tcp open|filtered postgresql

5900/tcp open|filtered vnc

6000/tcp open|filtered X11

6667/tcp open|filtered irc

8009/tcp open|filtered ajp13

8180/tcp open|filtered unknown

MAC Address: 08:00:27:AB:BD:5D (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 14.58 seconds

4.7. Service and Version Detection

The command sudo nmap -sV 10.0.0.6 was used to identify services and their reported software versions.

┌──(kali㉿Kali)-[~]

└─$ sudo nmap -sV 10.0.0.6

Starting Nmap 7.94 (https://nmap.org) at 2026-10-21 01:10 UTC

Nmap scan report for 10.0.0.6

Host is up (0.00023s latency).

Not shown: 977 closed tcp ports (reset)

PORT STATE SERVICE VERSION

21/tcp open ftp vsftpd 2.3.4

22/tcp open ssh OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)

23/tcp open telnet Linux telnetd

25/tcp open smtp Postfix smtpd

53/tcp open domain ISC BIND 9.4.2

80/tcp open http Apache httpd 2.2.8 ((Ubuntu) DAV/2)

111/tcp open rpcbind 2 (RPC #100000)

139/tcp open netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)

445/tcp open netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)

512/tcp open exec netkit-rsh rexecd

513/tcp open login?

514/tcp open shell Netkit rshd

1099/tcp open java-rmi GNU Classpath grmiregistry

1524/tcp open bindshell Metasploitable root shell

2049/tcp open nfs 2-4 (RPC #100003)

2121/tcp open ftp ProFTPD 1.3.1

3306/tcp open mysql MySQL 5.0.51a-3ubuntu5

5432/tcp open postgresql PostgreSQL DB 8.3.0 - 8.3.7

5900/tcp open vnc VNC (protocol 3.3)

6000/tcp open X11 (access denied)

6667/tcp open irc UnrealIRCd

8009/tcp open ajp13 Apache Jserv (Protocol v1.3)

8180/tcp open http Apache Tomcat/Coyote JSP engine 1.1

MAC Address: 08:00:27:AB:BD:5D (Oracle VirtualBox virtual NIC)

Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .

Nmap done: 1 IP address (1 host up) scanned in 67.01 seconds

4.8. Operating-System Detection

The command sudo nmap -O 10.0.0.6 was used to estimate the target operating system based on network responses.

┌──(kali㉿Kali)-[~]

└─$ sudo nmap -O 10.0.0.6

Starting Nmap 7.94 (https://nmap.org) at 2026-10-21 01:18 UTC

Nmap scan report for 10.0.0.6

Host is up (0.00070s latency).

Not shown: 977 closed tcp ports (reset)

PORT STATE SERVICE

21/tcp open ftp

22/tcp open ssh

23/tcp open telnet

25/tcp open smtp

53/tcp open domain

80/tcp open http

111/tcp open rpcbind

139/tcp open netbios-ssn

445/tcp open microsoft-ds

512/tcp open exec

513/tcp open login

514/tcp open shell

1099/tcp open rmiregistry

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2049/tcp open nfs

2121/tcp open ccproxy-ftp

3306/tcp open mysql

5432/tcp open postgresql

5900/tcp open vnc

6000/tcp open X11

6667/tcp open irc

8009/tcp open ajp13

8180/tcp open unknown

MAC Address: 08:00:27:AB:BD:5D (Oracle VirtualBox virtual NIC)

Device type: general purpose

Running: Linux 2.6.X

OS CPE: cpe:/o:linux:linux_kernel:2.6

OS details: Linux 2.6.9 - 2.6.33

Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .

Nmap done: 1 IP address (1 host up) scanned in 14.81 seconds

4.9. SMB User Enumeration

The command nmap --script smb-enum-users.nse 10.0.0.6 was used to enumerate user-account information exposed through SMB.

┌──(kali㉿Kali)-[~]

└─$ nmap --script smb-enum-users.nse 10.0.0.6

Starting Nmap 7.94 (https://nmap.org) at 2026-10-21 01:20 UTC

Nmap scan report for 10.0.0.6

Host is up (0.00082s latency).

Not shown: 977 closed tcp ports (conn-refused)

PORT STATE SERVICE

21/tcp open ftp

22/tcp open ssh

23/tcp open telnet

25/tcp open smtp

53/tcp open domain

80/tcp open http

111/tcp open rpcbind

139/tcp open netbios-ssn

445/tcp open microsoft-ds

512/tcp open exec

513/tcp open login

514/tcp open shell

1099/tcp open rmiregistry

1524/tcp open ingreslock

2049/tcp open nfs

2121/tcp open ccproxy-ftp

3306/tcp open mysql

5432/tcp open postgresql

5900/tcp open vnc

6000/tcp open X11

6667/tcp open irc

8009/tcp open ajp13

8180/tcp open unknown

Host script results:

| smb-enum-users:

| METASPLOITABLE\backup (RID: 1068)

| Full name: backup

| Flags: Account disabled, Normal user account

| METASPLOITABLE\bin (RID: 1004)

| Full name: bin

| Flags: Account disabled, Normal user account

| METASPLOITABLE\bind (RID: 1210)

| Flags: Account disabled, Normal user account

| METASPLOITABLE\daemon (RID: 1002)

| Full name: daemon

| Flags: Account disabled, Normal user account

| METASPLOITABLE\dhcp (RID: 1202)

| Flags: Account disabled, Normal user account

| METASPLOITABLE\distccd (RID: 1222)

| Flags: Account disabled, Normal user account

| METASPLOITABLE\ftp (RID: 1214)

| Flags: Account disabled, Normal user account

| METASPLOITABLE\games (RID: 1010)

| Full name: games

| Flags: Account disabled, Normal user account

| METASPLOITABLE\gnats (RID: 1082)

| Full name: Gnats Bug-Reporting System (admin)

| Flags: Account disabled, Normal user account

| METASPLOITABLE\irc (RID: 1078)

| Full name: ircd

| Flags: Account disabled, Normal user account

| METASPLOITABLE\klog (RID: 1206)

| Flags: Account disabled, Normal user account

| METASPLOITABLE\libuuid (RID: 1200)

| Flags: Account disabled, Normal user account

| METASPLOITABLE\list (RID: 1076)

| Full name: Mailing List Manager

| Flags: Account disabled, Normal user account

| METASPLOITABLE\lp (RID: 1014)

| Full name: lp

| Flags: Account disabled, Normal user account

| METASPLOITABLE\mail (RID: 1016)

| Full name: mail

| Flags: Account disabled, Normal user account

| METASPLOITABLE\man (RID: 1012)

| Full name: man

| Flags: Account disabled, Normal user account

| METASPLOITABLE\msfadmin (RID: 3000)

| Full name: msfadmin,,,

| Flags: Normal user account

| METASPLOITABLE\mysql (RID: 1218)

| Full name: MySQL Server,,,

| Flags: Account disabled, Normal user account

| METASPLOITABLE\news (RID: 1018)

| Full name: news

| Flags: Account disabled, Normal user account

| METASPLOITABLE\nobody (RID: 501)

| Full name: nobody

| Flags: Account disabled, Normal user account

| METASPLOITABLE\postfix (RID: 1212)

| Flags: Account disabled, Normal user account

| METASPLOITABLE\postgres (RID: 1216)

| Full name: PostgreSQL administrator,,,

| Flags: Account disabled, Normal user account

| METASPLOITABLE\proftpd (RID: 1226)

| Flags: Account disabled, Normal user account

| METASPLOITABLE\proxy (RID: 1026)

| Full name: proxy

| Flags: Account disabled, Normal user account

| METASPLOITABLE\root (RID: 1000)

| Full name: root

| Flags: Account disabled, Normal user account

| METASPLOITABLE\service (RID: 3004)

| Full name: ,,,

| Flags: Account disabled, Normal user account

| METASPLOITABLE\sshd (RID: 1208)

| Flags: Account disabled, Normal user account

| METASPLOITABLE\sync (RID: 1008)

| Full name: sync

| Flags: Account disabled, Normal user account

| METASPLOITABLE\sys (RID: 1006)

| Full name: sys

| Flags: Account disabled, Normal user account

| METASPLOITABLE\syslog (RID: 1204)

| Flags: Account disabled, Normal user account

| METASPLOITABLE\telnetd (RID: 1224)

| Flags: Account disabled, Normal user account

| METASPLOITABLE\tomcat55 (RID: 1220)

| Flags: Account disabled, Normal user account

| METASPLOITABLE\user (RID: 3002)

| Full name: just a user,111,,

| Flags: Normal user account

| METASPLOITABLE\uucp (RID: 1020)

| Full name: uucp

| Flags: Account disabled, Normal user account

| METASPLOITABLE\www-data (RID: 1066)

| Full name: www-data

|_ Flags: Account disabled, Normal user account

Nmap done: 1 IP address (1 host up) scanned in 13.64 seconds

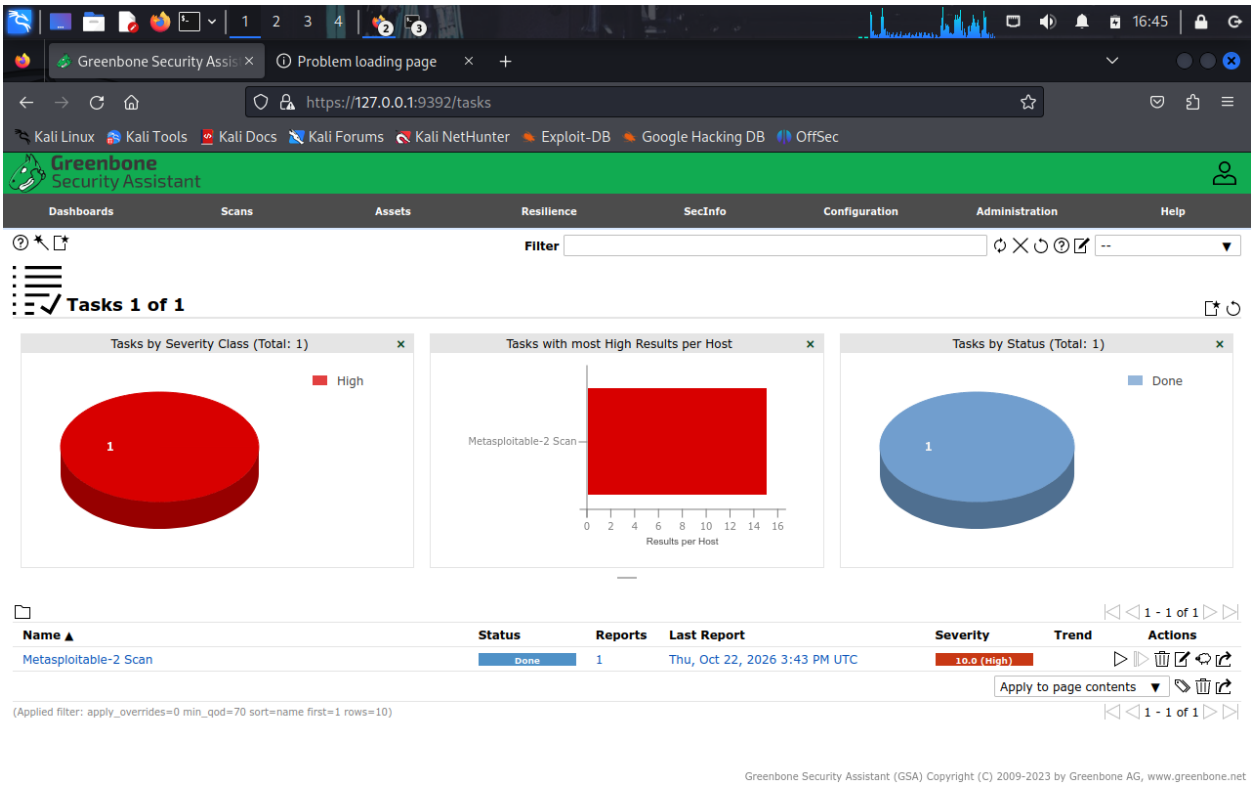
OpenVAS Vulnerability Report

5. OpenVAS Vulnerability Report

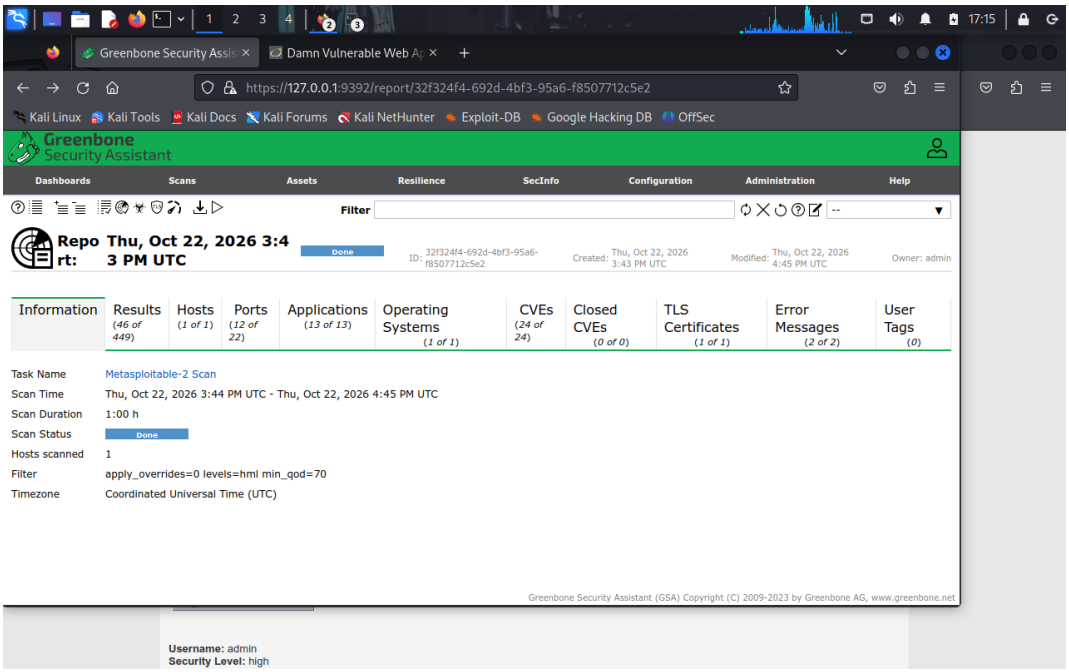
OpenVAS was used to review the vulnerability-assessment interface and its available report sections, including results, hosts, ports, applications, operating systems, CVEs, TLS certificates, and error messages.

The methodology was limited to the commands, outputs, and screenshots documented in the supplied evidence.

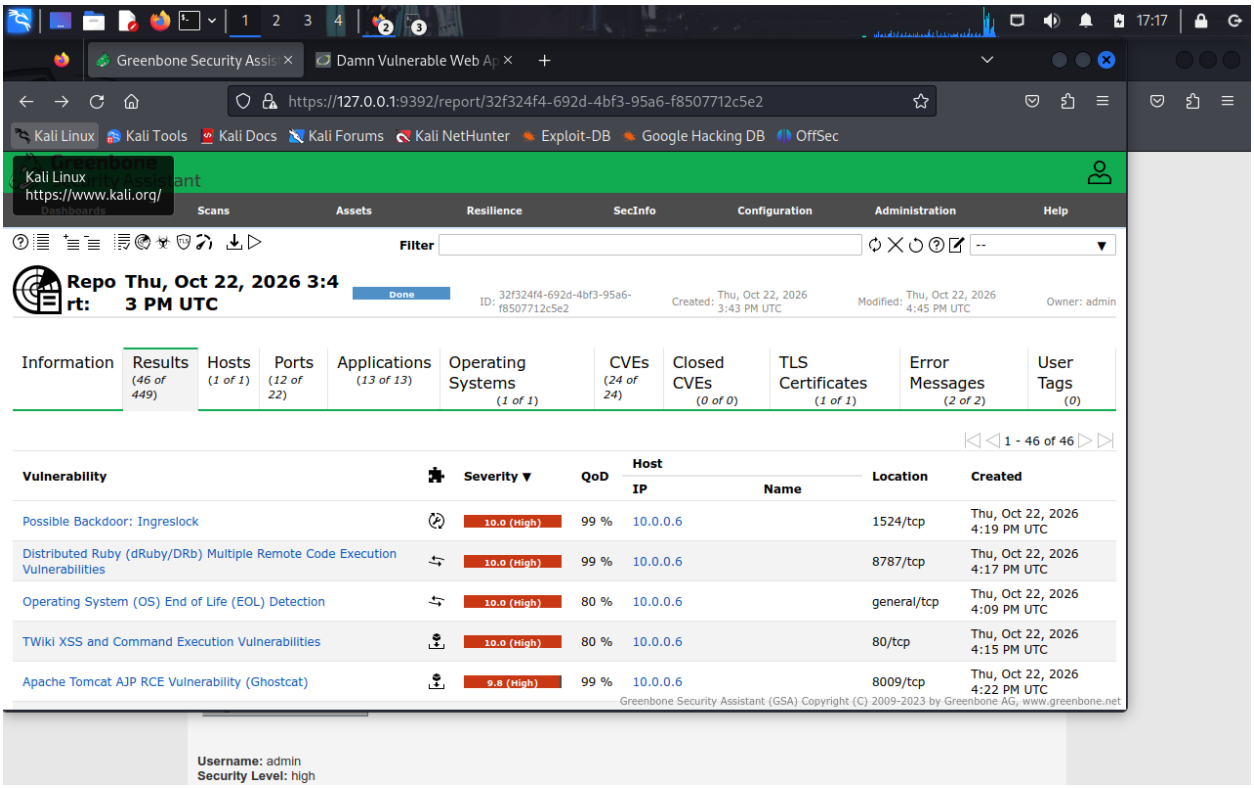
5.1. OpenVAS Main Page



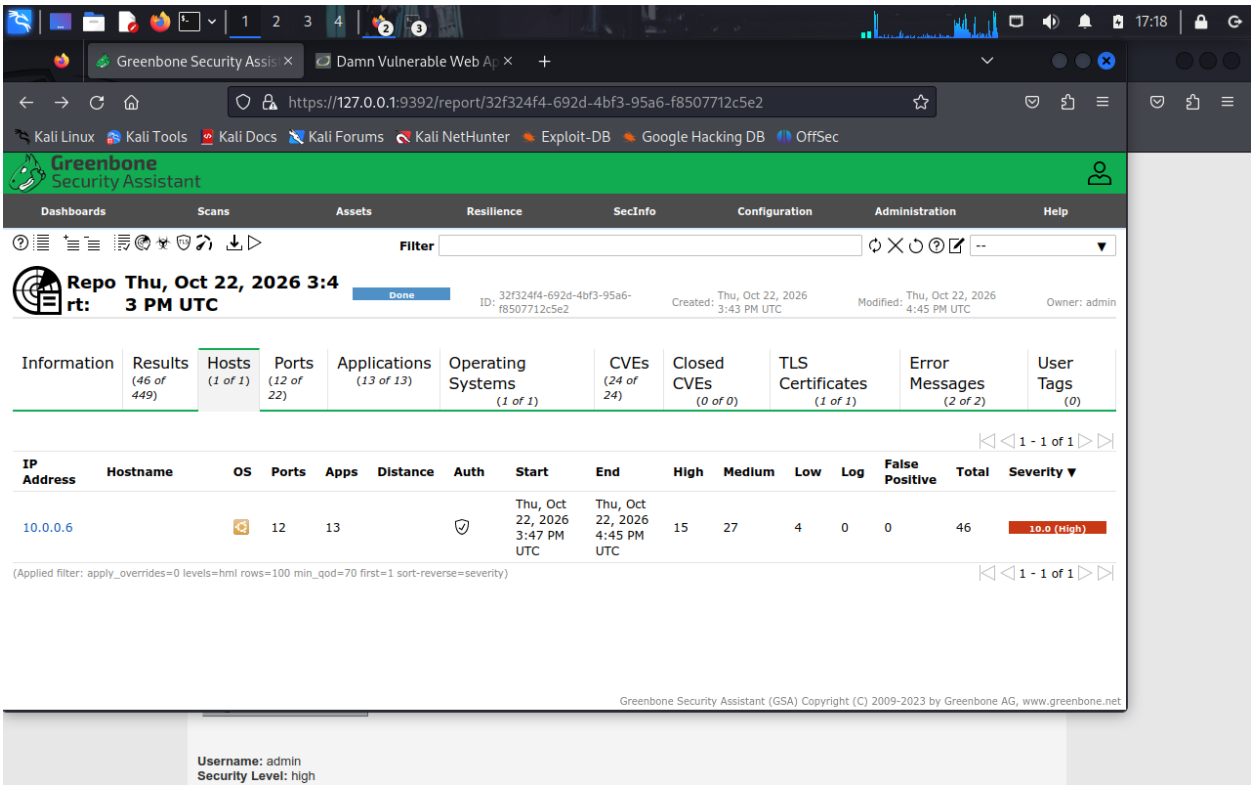
5.2. Information Tab



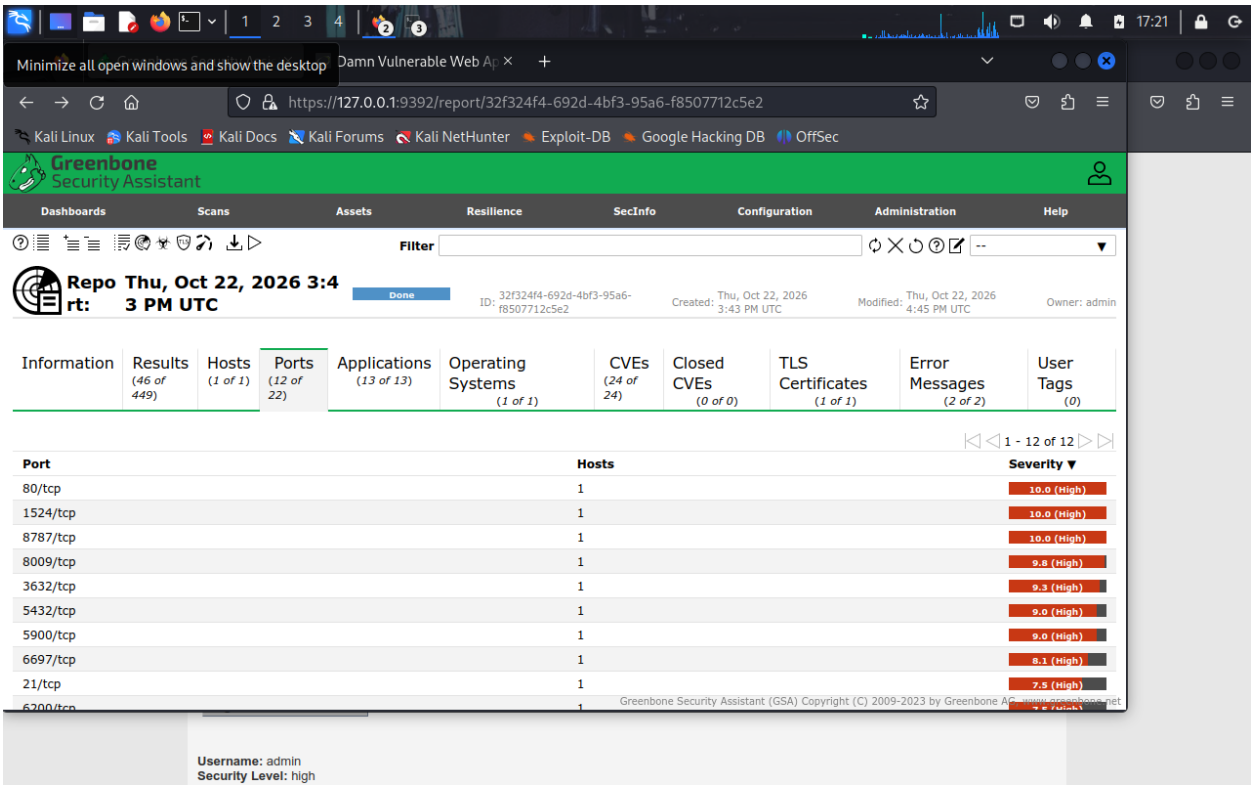
5.3. Result Tab



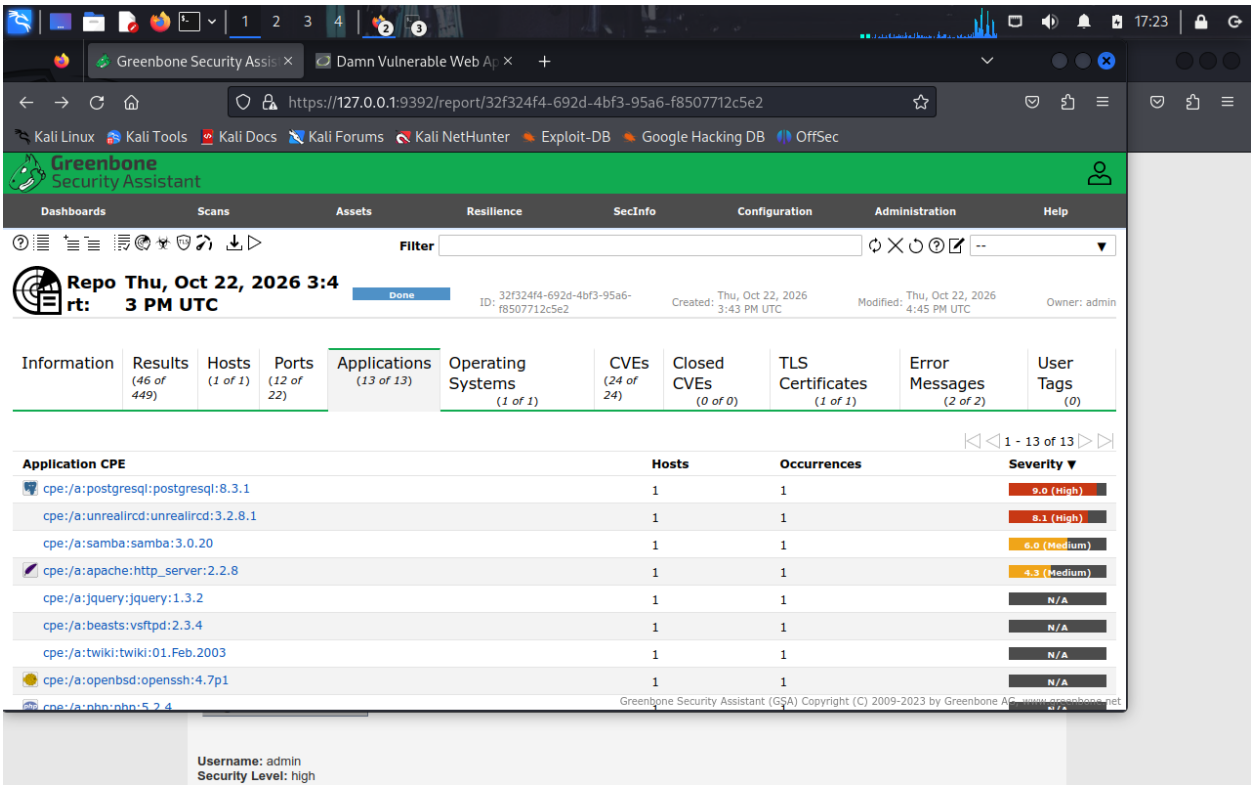
5.4. Host Tab


Greenbone Security Assistant
Dashboards
Scans
Assets
Resilience
SecInfo
Configuration
Administration
Help
Filter
Report: Thu, Oct 22, 2026 3:43 PM UTC
Done
Information
Results (46 of 449)
Hosts (1 of 1)
Ports (12 of 22)
Applications (13 of 13)
Operating Systems (1 of 1)
CVEs (24 of 24)
Closed CVEs (0 of 0)
TLS Certificates (1 of 1)
Error Messages (2 of 2)
User Tags (0)
IP Address
Hostname
OS
Ports
Apps
Distance
Auth
Start
End
High
Medium
Low
Log
False Positive
Total
Severity
10.0.0.6
12
13
Thu, Oct 22, 2026 3:47 PM UTC
Thu, Oct 22, 2026 4:45 PM UTC
15
27
4
0
0
46
10.0 (High)
Greenbone Security Assistant (GSA) Copyright (C) 2009-2023 by Greenbone AG, www.greenbone.net
Username: admin
Security Level: high

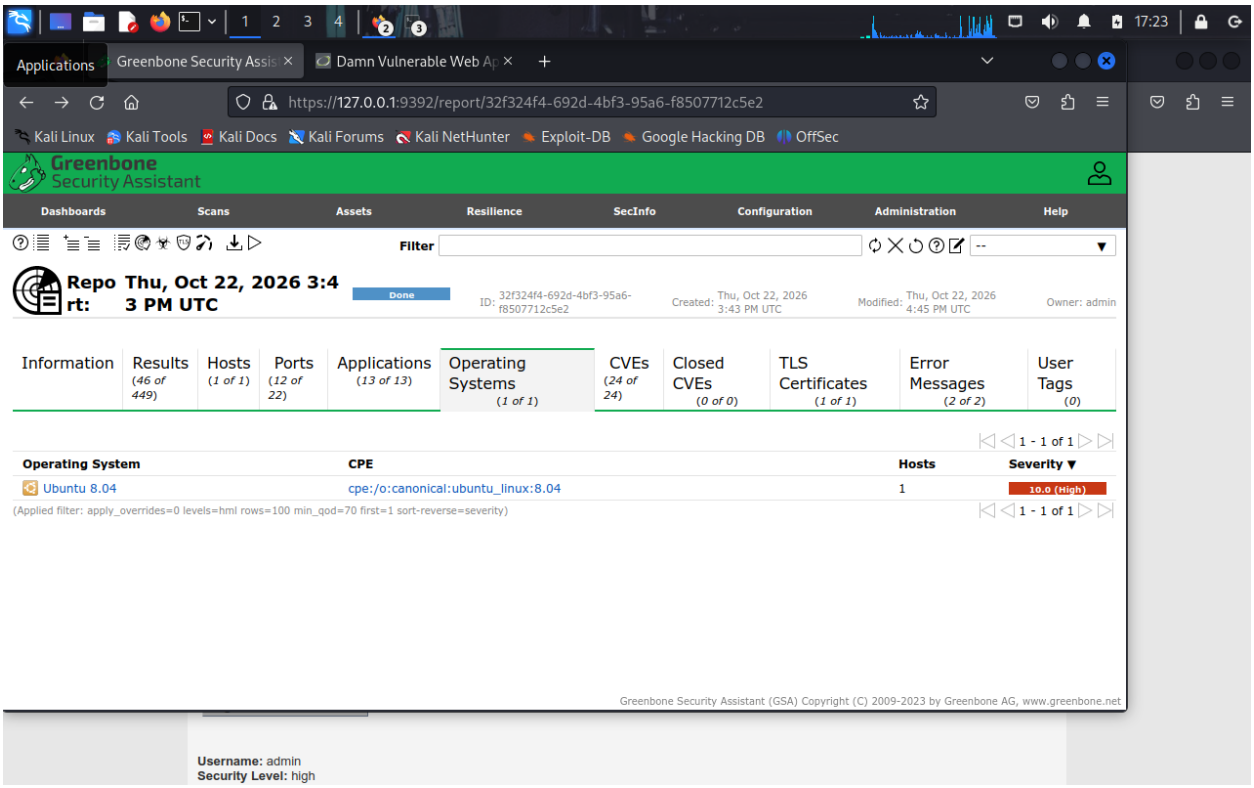
5.5. Ports Tab


Minimize all open windows and show the desktop
https://127.0.0.1:9392/report/32f324f4-692d-4bf3-95a6-f8507712c5e2
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
OffSec
Greenbone Security Assistant
Dashboards
Scans
Assets
Resilience
SecInfo
Configuration
Administration
Help
Filter
Report: Thu, Oct 22, 2026 3:43 PM UTC
Done
ID: 32f324f4-692d-4bf3-95a6-f8507712c5e2
Created: Thu, Oct 22, 2026 3:43 PM UTC
Modified: Thu, Oct 22, 2026 4:45 PM UTC
Owner: admin
Information
Results (46 of 449)
Hosts (1 of 1)
Ports (12 of 22)
Applications (13 of 13)
Operating Systems (1 of 1)
CVEs (24 of 24)
Closed CVEs (0 of 0)
TLS Certificates (1 of 1)
Error Messages (2 of 2)
User Tags (0)
1 - 12 of 12
Port
Hosts
Severity
80/tcp 1 10.0 (High)
1524/tcp 1 10.0 (High)
8787/tcp 1 10.0 (High)
8009/tcp 1 9.8 (High)
3632/tcp 1 9.3 (High)
5432/tcp 1 9.0 (High)
5900/tcp 1 9.0 (High)
6697/tcp 1 8.1 (High)
21/tcp 1 7.5 (High)
Greenbone Security Assistant (GSA) Copyright (C) 2009-2023 by Greenbone AG
Username: admin
Security Level: high

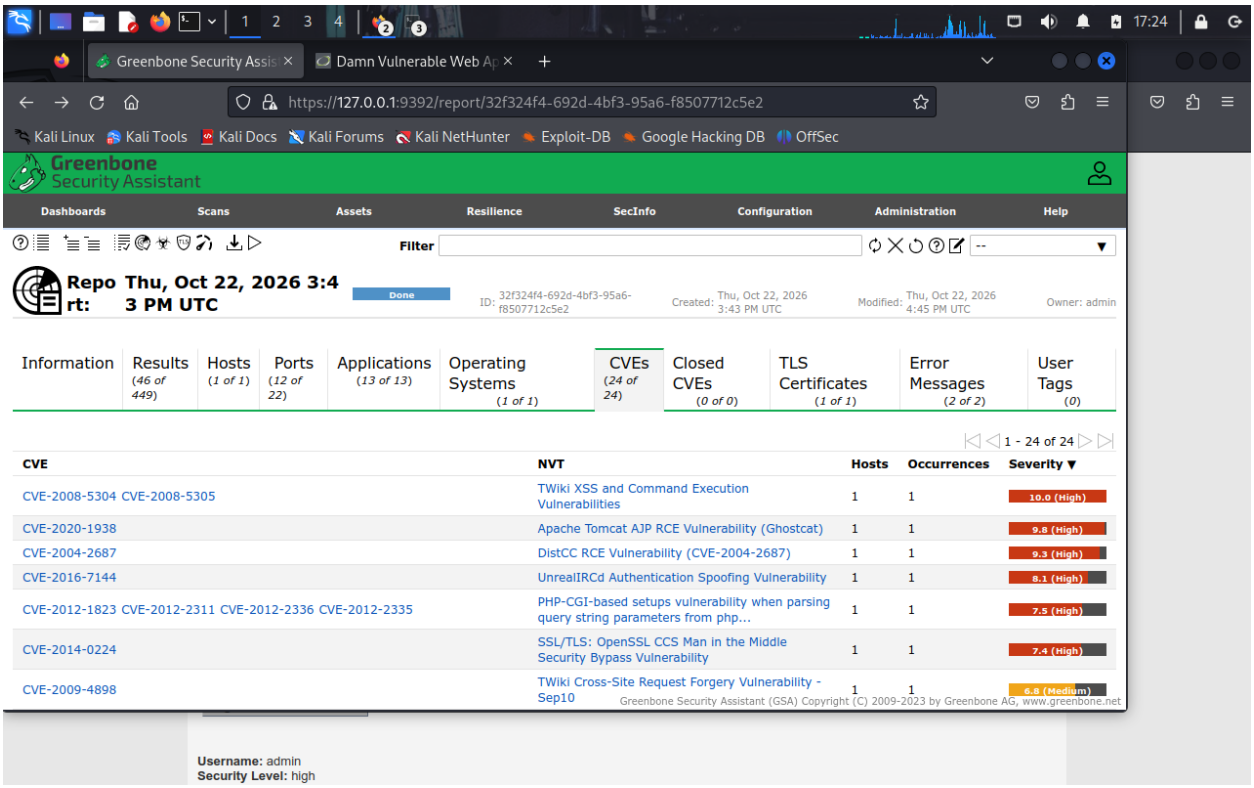
5.6. Applications Tab



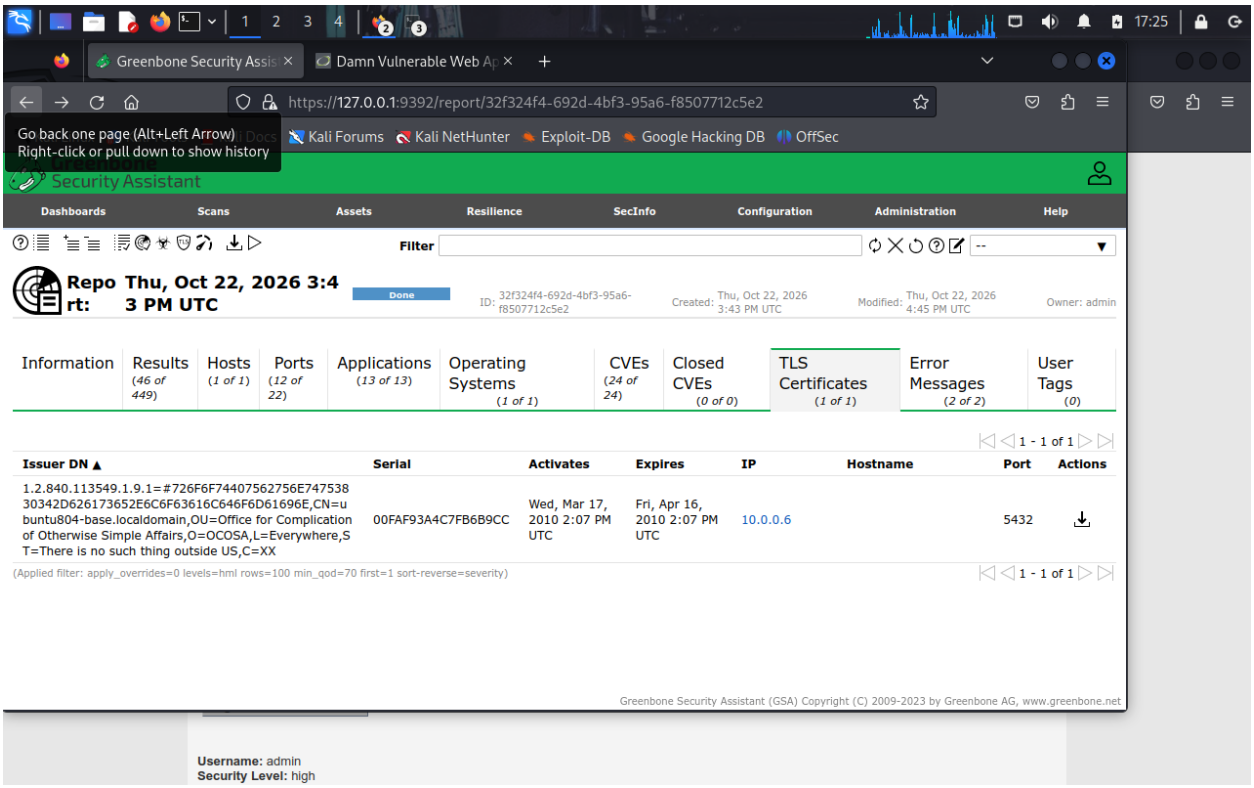
5.7. Operating System Tab



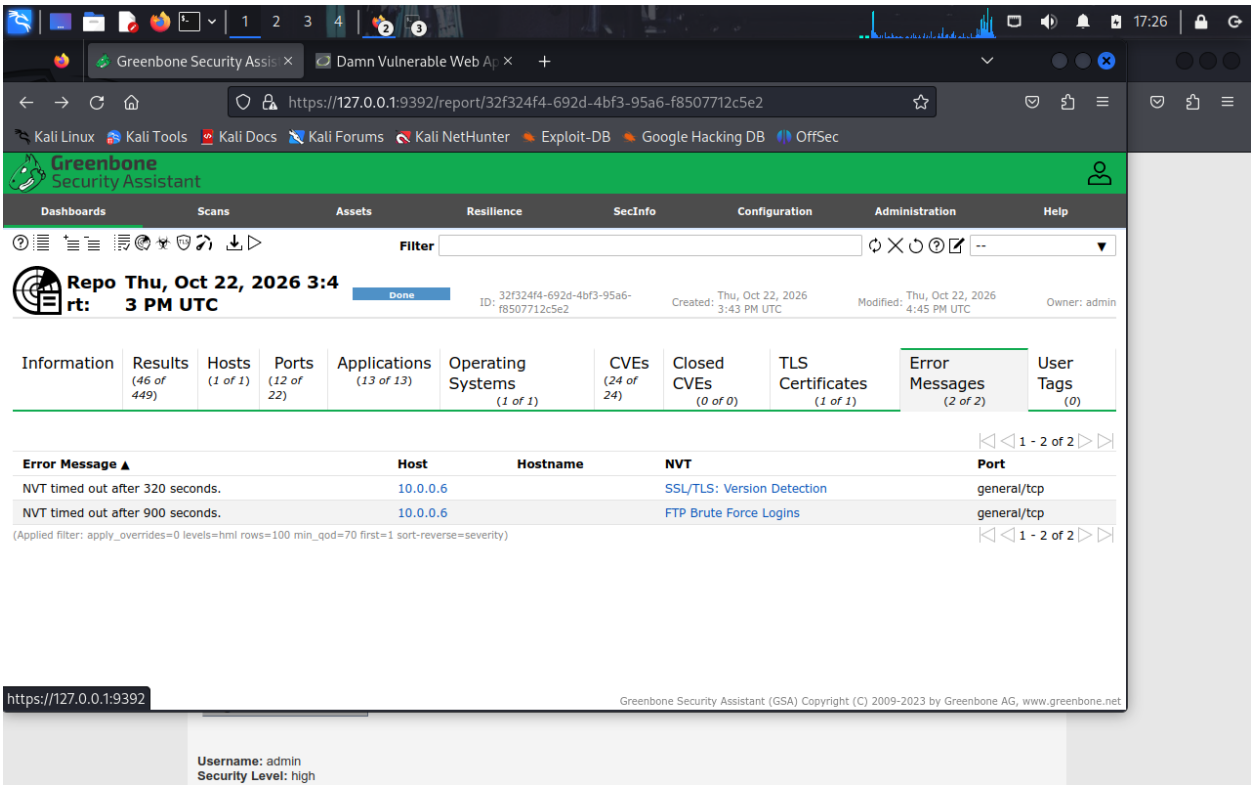
5.8. CVEs Tab


Greenbone Security Assistant
Dashboards
Scans
Assets
Resilience
SecInfo
Configuration
Administration
Help
Report: Thu, Oct 22, 2026 3:43 PM UTC
Done
Information
Results (46 of 449)
Hosts (1 of 1)
Ports (12 of 22)
Applications (13 of 13)
Operating Systems (1 of 1)
CVEs (24 of 24)
Closed CVEs (0 of 0)
TLS Certificates (1 of 1)
Error Messages (2 of 2)
User Tags (0)
1 - 24 of 24
CVE
NVT
Hosts
Occurrences
Severity
CVE-2008-5304 CVE-2008-5305
TWiki XSS and Command Execution Vulnerabilities
10.0 (High)
CVE-2020-1938
Apache Tomcat AJP RCE Vulnerability (Ghostcat)
9.8 (High)
CVE-2004-2687
DistCC RCE Vulnerability (CVE-2004-2687)
9.3 (High)
CVE-2016-7144
UnrealIRCd Authentication Spoofing Vulnerability
8.1 (High)
CVE-2012-1823 CVE-2012-2311 CVE-2012-2336 CVE-2012-2335
PHP-CGI-based setups vulnerability when parsing query string parameters from php...
7.5 (High)
CVE-2014-0224
SSL/TLS: OpenSSL CCS Man in the Middle Security Bypass Vulnerability
7.4 (High)
CVE-2009-4898
TWiki Cross-Site Request Forgery Vulnerability - Sep10
6.8 (Medium)
Greenbone Security Assistant (GSA) Copyright (C) 2009-2023 by Greenbone AG, www.greenbone.net
Username: admin
Security Level: high

5.9. TLS Certificate Tab



5.10. Error Message Tab



6. Conclusion

The assessment demonstrated how Nmap and OpenVAS can support network discovery, service enumeration, and vulnerability assessment in a controlled laboratory environment.

The Nmap results identified multiple open ports and exposed services on the Metasploitable target. Service detection revealed several legacy software versions, while operating-system detection suggested an older Linux kernel. SMB enumeration also disclosed user-account information. These observations demonstrate the importance of minimizing exposed services, maintaining supported software versions, and restricting unnecessary information disclosure.

Recommended next steps include validating potential vulnerabilities against reliable references, reviewing the actual OpenVAS findings and severity ratings, disabling unnecessary services, updating vulnerable software, restricting access to administrative and legacy protocols, and documenting remediation results.

This exercise was conducted in an authorized virtual lab for educational purposes. The findings should not be interpreted as confirmation of successful exploitation, and any proposed remediation should be based on validated assessment results.